Hydraulic characterization of ryegrass-fiber reinforced soil slopes under rainfall conditions

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Received 13 November 2025, accepted 28 March 2026

ABSTRACT: Rainfall-induced instability poses a critical threat to soil slopes. Although fiber reinforcement and vegetation are common stabilization methods, the hydraulic synergy within "root-fiber-soil" systems remains poorly understood. This study addresses the impact of fiber wettability by comparing hydrophilic coir fiber (CF) and hydrophobic polypropylene fiber (PP) combined with ryegrass. Indoor rainfall model tests were conducted to evaluate erosion resistance and hydraulic response. Results indicate a non-linear correlation between fiber content and hydraulic parameters. The optimal dosages for hydraulic regulation were identified as 0.5% for PP and 0.75% for CF. Under heavy rainfall (80 mm/h), the ryegrass-PP system exhibited superior drainage efficiency, reducing peak volumetric water content (VWC) by 10.8% relative to the ryegrass-CF system. Mechanistically, hydrophobic PP facilitates a "drainage synergy" with root channels, accelerating pore water pressure (PWP) dissipation. Conversely, hydrophilic CF induces a "sponge effect", resulting in increased moisture retention. Consequently, a wettability-based design criterion is proposed: hydrophobic fibers are preferred for reducing PWP in high-rainfall regions, whereas hydrophilic fibers are optimal for moisture retention in arid zones.

KEYWORDS: Geosynthetics, Indoor Rainfall Test, Ryegrass, Coir Fiber, Polypropylene Fiber, Hydraulic Properties, Clays

REFERENCE: Li, L., Yuan, X., Li, Y., Zhu, S., Zong, Y. and Zhang, D. (2026). Hydraulic characterization of ryegrass-fiber reinforced soil slopes under rainfall conditions. *Geosynthetics International.* [https://doi.org/10.1680/jgein.25.00203]

1. INTRODUCTION

In the context of global climate change, the increasing frequency of extreme rainfall events has significantly exacerbated the risk of slope instability, posing severe threats to infrastructure and human safety. Consequently, soil reinforcement utilizing fibrous materials has been extensively investigated as a sustainable technique for mitigating these geo-hazards and enhancing geotechnical properties. Natural fibers, such as coir fiber, have demonstrated significant efficacy in improving the shear strength, stress-strain behavior, and slope stability of red clay and reconstituted fine-grained soils by providing distributed tension resistance and restricting soil particle displacement (Dasaka and Sumesh 2011; Guo 2023; Parmar and Shukla 2026). Similarly, synthetic fibers, such as polypropylene (PP), have proven effective in enhancing the compressive strength, tensile performance, and erosion durability of loess, cement-stabilized soils, and lightweight aggregates (Chen *et al.* 2022; Yang *et al.* 2022; Liu *et al.* 2024; Khalid *et al.* 2025). Advanced analytical methods, including micro-numerical simulations and dynamic fractal analysis, have further elucidated the mechanical reinforcement mechanisms of these fiber-reinforced composites, revealing the critical role of the "bridging effect" and interfacial friction in arresting crack propagation (Yang *et al.* 2021; Li *et al.* 2024a). Moreover, recent research has expanded into complex stabilization strategies—such as coupling fibers with chemical additives (e.g. lime, fly ash, nano-clay) or biopolymers (e.g. xanthan gum) (Tiwari and Satyam 2021; Jiang *et al.* 2022a; Lu *et al.* 2022; Feng *et al.* 2023; Li *et al.* 2024b)—and investigating their durability under extreme conditions, including freeze-thaw cycles and seawater corrosion (Chen *et al.* 2024; Saroj *et al.* 2024; Ma *et al.* 2025; Rajaee *et al.* 2025). Despite these advancements, existing literature has predominantly focused on mechanical enhancement under static or dynamic loading. There remains a critical lack of comparative analysis regarding the hydrological response of these fibers. Specifically, the fundamental interaction between fiber surface properties and soil moisture migration remains a "black box." It is unclear how their distinct surface wettability (hydrophilic coir fiber vs. hydrophobic PP fiber) regulates soil permeability and pore water pressure (PWP) under dynamic rainfall conditions, which is a governing factor in rainfall-induced landslides.

In the domain of ecological engineering, vegetation is recognized as a pivotal agent for slope stabilization due to its ecosystem services and aesthetic value. Field trials and numerical studies have quantified the contribution of root density to matric suction and slope stability (Li *et al.* 2022; Hai *et al.* 2023; Wang *et al.* 2024; Zhang *et al.* 2025), while also analyzing the coupled effects of root architecture and slope gradient on rainfall infiltration (Dyson *et al.* 2023; Huang *et al.* 2023; Tai *et al.* 2023, 2025). However, the efficacy of vegetation is often limited during the early growth stages, leaving slopes vulnerable to erosion before a dense root network is established. Although the synergistic application of vegetation with geosynthetics (e.g. geotextiles, geocells) has been explored for slope restoration in semi-arid regions (Shao *et al.* 2014; Sheela *et al.* 2019; Broda *et al.* 2020; Zeng *et al.* 2023; Kim *et al.* 2024a; Li *et al.* 2024c; Hajiazizi *et al.* 2025; Li *et al.* 2026), and recent reviews have summarized plant-based soil improvement schemes (Li *et al.* 2025a), the interaction mechanisms within the "root-fiber-soil" ternary composite system remain underexplored. Conceptually, fibers can provide immediate mechanical reinforcement, while roots offer long-term stabilization, forming a temporally complementary system. Yet, from a hydraulic perspective, few studies have differentiated the synergistic effects of plant roots with hydrophilic natural fibers versus hydrophobic synthetic fibers in regulating the unsaturated seepage field (Li *et al.* 2023). Understanding this interaction is crucial, as the water retention of hydrophilic fibers might aid plant survival but increase slope weight, whereas hydrophobic fibers might accelerate drainage but reduce drought resilience.

To address these knowledge gaps, this study conducts a comparative investigation into the hydraulic behavior of slopes reinforced with hydrophilic CF and hydrophobic PP combined with ryegrass. Distinct from traditional studies that focus separately on fiber mechanics (Patel and Singh 2020; Wu *et al.* 2023) or root reinforcement (Ni *et al.* 2023; Chakravarthy *et al.* 2025), this research integrates soil mechanics, hydrology, and plant physiology to answer the following key questions: (1) How do the contrasting wettability characteristics of coir and PP differentially affect volumetric water content (VWC) and PWP accumulation under varying rainfall intensities? (2) Is there a synergistic or antagonistic effect between fiber networks and root systems in modulating slope infiltration? and (3) What are the optimal fiber dosages to balance hydraulic regulation with mechanical reinforcement? By elucidating the non-linear hydraulic response mechanisms of these composite systems, this study provides theoretical insights for optimizing ecological slope engineering in regions susceptible to heavy rainfall, aiming to propose a wettability-based criterion for selecting reinforcement materials (Saurabh and Lal 2022).

2. TEST MATERIALS

The soil was taken from a pit 1 to 2 meters below the surface in Wuhan, China. Geotechnical tests were conducted in accordance with the *Highway Geotechnical Test Specification* (*JTG3430-2020*) (JTG 3430-2020 2020). A sieve test was conducted on the soil, using sieves with aperture diameters of 10 mm, 5 mm, 2 mm, 1 mm, 0.5 mm, 0.25 mm, and 0.075 mm, respectively. The particle grading curve is shown in Figure 1, which shows that Soil $Cu = 5.67 > 5$ and $3 > CC = 1.06 > 1$. Tests yielded a liquid limit of 33.7%, a plastic limit of 15.9%, and a plasticity index of 17.8. According to the Standard for Engineering Classification of Soil (GB/T 50145-2007) of China, the soil is classified as clay. Standard compaction tests were performed on the soil samples at various water contents. Table 1 shows the

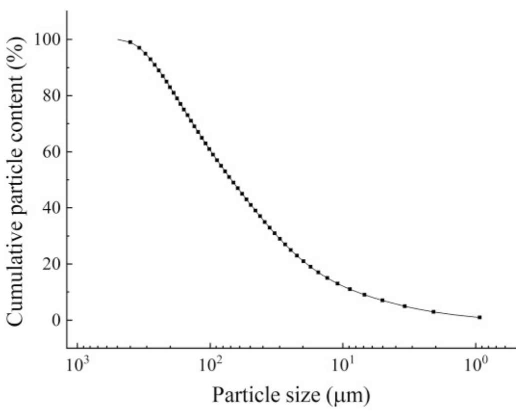

Figure 1. Particle grading curve

Table 1. Soil physical parameters

Sample	Maximum dry density (g/cm^{-3})	Optimum moisture content (%)	Liquid limit (%)	Plastic limit (%)	Plasticity index (%)
Soil	1.82	16.5	33.7	15.9	17.8

physical parameters of the soil obtained from the compaction test.

The basic physical parameters of the coir fibers and polypropylene fibers are summarized in Table 2. The CF appear pale yellow, with a length of 0.020∼0.025 m (Babu and Vasudevan 2008) and a circular cross-sectional shape. Chemically, they are composed of cellulose, hemicellulose, lignin, and pectin. Characterized by high cellulose and low hemicellulose content, CF demonstrate superior resistance to moisture and heat, as well as excellent mechanical properties. Furthermore, they are easily accessible, cost-effective, and environmentally friendly. When mixed with soil, these fibers resist agglomeration and disperse readily, ensuring good homogeneity within the soil matrix. The PP selected for this study have a length of 0.015 m (Tang *et al.* 2012) and appear as silver-white bundled monofilaments, with individual fibers being transparent. These fibers are characterized by their low density, superior tensile properties, and strong resistance to acid and alkali corrosion. Although coir and PP are currently less common in practical engineering applications, they have been identified as key materials for future development. Given the critical role of fibers in soil improvement and their immense potential in roadbed and slope engineering, investigating the performance of fiber-reinforced soil under rainfall conditions holds significant practical engineering value.

3. EXPERIMENTAL METHODS

The test model mainly consists of three parts: slope model, rainfall simulation device, and collection device (Figure 2). The slope model consists of two kinds of steel frames with 15° and 35° angles. A model box with dimensions of 1000 mm × 350 mm × 150 mm (length × width × height) and a plate thickness of 2 mm was placed on a steel frame. The simulated rainfall device consists of a water storage tank, a water pump, a rainfall sprinkler, a rainfall bracket, and a model box bracket. The collection device consists of a rainwater storage tank and a runoff collection box. The rainfall device consists of a water storage tank, a water pump, a rainfall nozzle, a rainfall bracket, and a model box bracket. The collection device consists of the rainwater storage tank and the runoff collection box. Six pore water pressure (PWP) sensors are used in this test, with a range of 0∼5 kPa, accuracy ≤0.5% FS (full-scale linear error), 3 VWC sensors, with a range of 0∼100%, and an accuracy of within ± 3%. The spatial distribution of the rainfall simulation apparatus was evaluated using rain gauges arranged in a grid pattern, yielding a calculated Christiansen Uniformity Coefficient (CUC) consistently exceeding 0.85. Additionally, the stain method (using filter paper) was employed to characterize the microphysical properties of the raindrops, resulting in a measured median drop diameter of approximately 1.8 mm. The data acquisition system comprises six PWP sensors (range: 0∼5 kPa; accuracy: ≤ ±0.5% FS) and three VWC sensors (range: 0∼100%; accuracy: ±3%). Monitoring points were established along the longitudinal axis of the model box at three representative sections: the slope toe, middle slope, and slope crest, located at distances of 50 mm, 500 mm, and 950 mm from the front end of the model box, respectively. At each section, a VWC sensor was embedded vertically at a depth of 50 mm below the slope surface, while the PWP sensors were arranged in pairs and installed at depths of 3 mm and 12 mm below the surface.

Table 2. Physical properties of materials

Fiber material	CF	PP
Lengths (m)	0.020∼0.025	0.015
Tensile strength (MPa)	85.20∼110.50	>486
Modulus of elasticity (MPa)	2110∼2350	>4000

The reinforced soil with different dosages of fibers will be filled into the model box in three layers, with a filling height of 50 mm in each layer. The moisture sensors (which measure VWC) are buried in three layers, and the burying depth for each layer is 50 mm. The PWP sensors were laid in three layers with burial depths of 3 mm and 12 mm. The specific layout of the moisture sensor and PWP sensor is shown in Figure 2(b). After 30 days of maintenance, the model was placed on a steel frame with a slope of 15° or 35° for the rainfall test.

The rainfall tests were conducted on the slope models containing different quantities of fibers. The performance of fiber-vegetation reinforced slopes was investigated by planting ryegrass, varying slope gradient, and rainfall intensity. The ryegrass selected for this study was subjected to 5 months growth and consolidation period. Before testing, the root systems were fully developed, forming an extensive mechanical interlocking network within the soil matrix. This ensured the full realization of the roots' soil reinforcement and hydrological regulation capabilities. Given the destructive nature of the experiments in this study, physical replication on the same experimental group was not performed. However, rigorous preliminary experiments demonstrated that the coefficient of variation (CV) for key parameters was less than 10%, indicating good reproducibility under the standardized protocol. Furthermore, all comparative experiments were conducted during the same period, under identical

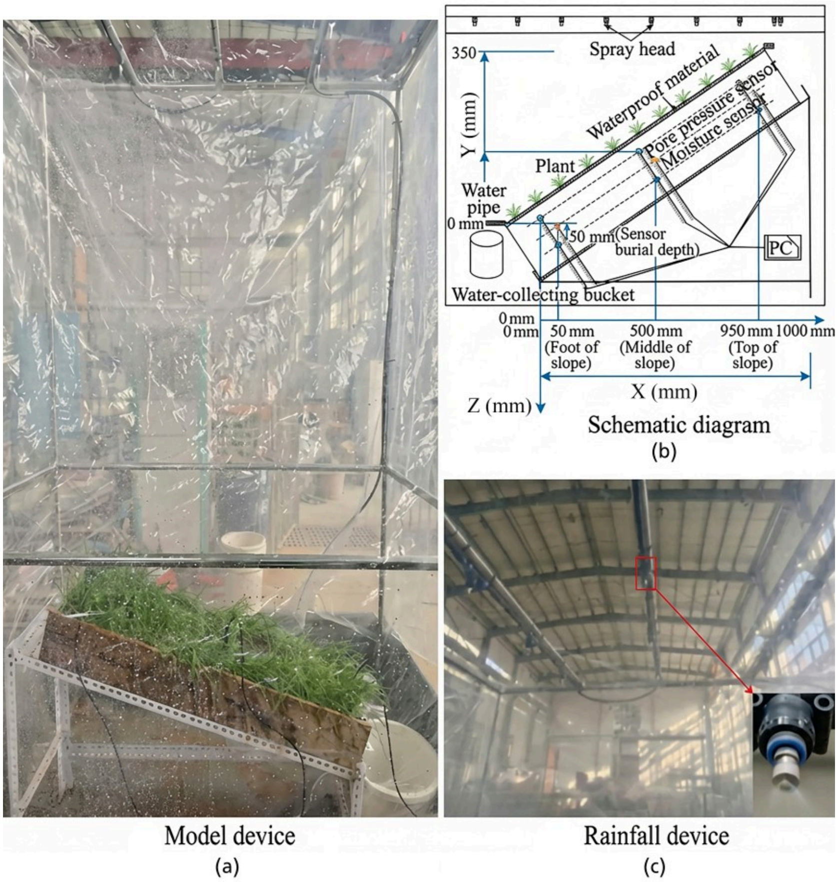

Figure 2. Test model and sensor arrangement

environmental conditions, and following the same standardized procedures to ensure the minimization of systematic errors. The specific test program is shown in Table 3 below.

To ensure the uniform dispersion of fibers within the soil matrix and prevent "balling" or agglomeration (particularly with the longer CF), a "dry-mixing followed by incremental addition" technique was adopted. First, the air-dried soil was pulverized and passed through a 2 mm sieve. Before water addition, the pre-weighed fibers were manually teased apart into a loose state and added to the dry soil in 3∼5 small increments. After each addition, the mixture was thoroughly stirred by hand until the individual fiber filaments were visually observed to be randomly distributed in three dimensions without distinct clustering. Subsequently, water was sprayed evenly onto the dry mixture to reach the optimum moisture content (16.5%). The prepared moist soil was then sealed in plastic bags and allowed to equilibrate for 24 h to ensure uniform moisture distribution within the soil-fiber mixture.

The construction of the slope model employed a layer-wise compaction method using the "mass-volume control" technique to strictly ensure that the dry density of each layer reached the target value (1.82 g/cm^3, corresponding to a compaction degree of 95%). The model was compacted in three layers along the height of the box, with a compacted thickness of 50 mm per layer. Prior to placing the subsequent layer, the surface of the compacted layer was scarified (roughened) to a depth of approximately 2∼3 mm using a scraper. This procedure was critical to eliminate weak interfaces between layers and ensure the hydraulic continuity and mechanical integrity of the slope model during rainfall infiltration and deformation.

Table 3. Pilot program

Reinforcing material	Reinforcing material mixing/%	Slope gradient /°	Rainfall/ mm·h^{-1}
	0	15/35	50/80/120
PP	0.3	15/35	50/80/120
	0.5	15/35	50/80/120
	0.7	15/35	50/80/120
CF	0.5	15/35	50/80/120
	0.75	15/35	50/80/120
	1	15/35	50/80/120

4. RESULTS AND DISCUSSION

4.1. Influence of CF on the hydraulic characteristics of slopes

CF can effectively regulate PWP in fiber-reinforced soil slopes, enhance the effective stress of the soil mass, and improve slope stability (Li *et al*. 2025b). Figure 3 illustrates the variation in the peak VWC of CF-reinforced soil slopes with increasing fiber content under rainfall

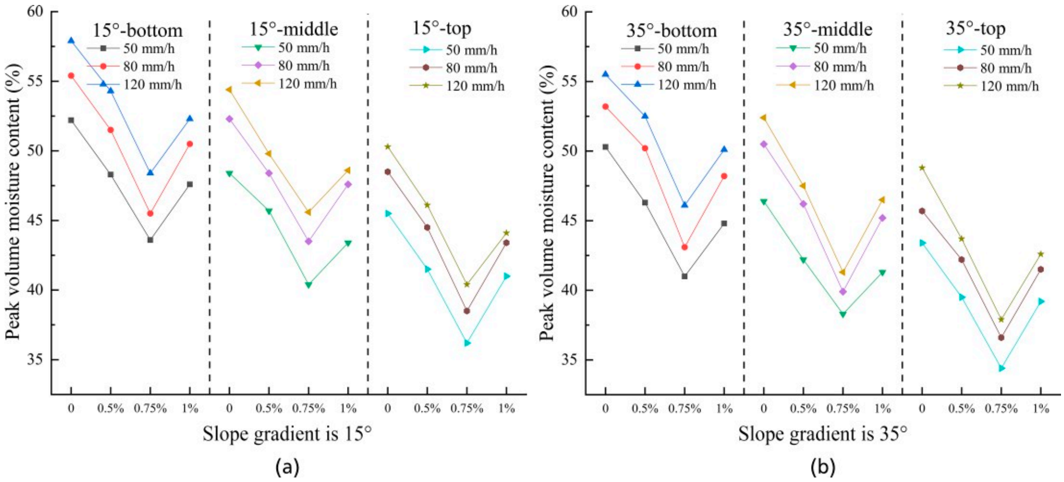

Figure 3. Variation in VWC of slope mass at different CF

conditions. At a slope angle of 15° and a rainfall intensity of 50 mm/h, the VWC at the toe of slopes reinforced with 0.5%, 0.75%, and 1% CF decreased by 7.5%, 16.5%, and 8.8%, respectively, relative to the unreinforced slope. This indicates that the unreinforced soil slope possesses a relatively poor pore structure and low permeability, leading to the accumulation of rainwater within the slope and consequently higher VWC. The incorporation of CF ameliorates the pore structure, facilitating the rapid drainage of rainwater; consequently, the peak VWC of the fiber-reinforced slopes is lower than that of the unreinforced slope. At a fiber content of 0.75%, the three-dimensional reticulated network formed by the fibers within the soil matrix maximizes the slope strength. At this optimum content, the synergistic effect between the CF and soil particles is most significant, yielding the optimal reduction in slope VWC. When the CF content is below 0.75%, the resulting three-dimensional network is relatively sparse, offering only limited improvement to the pore structure. In this phase, the fibers primarily act to obstruct soil pores, slightly retarding rainwater infiltration and thereby reducing the VWC. However, when the fiber content exceeds 0.75%, although the internal fiber network becomes denser, the formation of excessive interconnected pores within the slope structure compromises its mechanical strength. This phenomenon accelerates internal moisture migration, causing the VWC to rise with increasing porosity, which ultimately diminishes slope stability.

Rainfall intensity serves as a critical external hydraulic boundary governing the evolution of the unsaturated seepage field. As illustrated in Figure 4, increasing the rainfall intensity from 50 mm/h to 120 mm/h significantly amplifies the hydraulic gradient at the soil-atmosphere interface, thereby accelerating the wetting front advancement and shortening the time required for the soil to reach quasi-saturation. Spatially, driven by the gravitational potential field, moisture migrates downward, establishing a general VWC gradient of "slope bottom > slope middle slope > slope top". However, the sensitivity of these locations to rainfall intensity changes—manifested as the peak VWC increment—exhibits a distinct non-uniform trend of "slope middle > slope top > slope bottom". Quantitative analysis of the 0.5% CF-reinforced slope (15° gradient) reveals that under 120 mm/h rainfall, the peak VWC increment in the slope middle reaches 13.2%, surpassing that of the slope top (11.9%) and the slope bottom (11%). This phenomenon is mechanistically attributed to the "run-on effect": the slope middle acts as a transitional accumulation zone, receiving both direct precipitation and surface runoff/interflow from the high-potential crest, which maximizes its moisture flux sensitivity. Conversely, the slope bottom, despite having the lowest gravitational potential and serving as the ultimate drainage sink, maintains a near-saturation state even under lower intensities; consequently, its available pore space to buffer additional moisture flux is limited, resulting in the lowest marginal response to intensified rainfall.

Figure 5 illustrates variations in peak PWP within reinforced soil slopes under rainfall conditions at different CF content levels. Experimental data indicate that peak PWP values in fiber-reinforced slopes are generally lower than those in unreinforced plain soil slopes, exhibiting a trend of "initial decrease followed by a rebound" as fiber content increases, with 0.75% identified as the critical optimal content for PWP control. For instance, under a rainfall intensity of 50 mm/h and a 15° slope, the peak PWP at the lower layer of the slope toe significantly decreased from 4.44 kPa in plain soil to 3.65 kPa at 0.75% fiber content, representing a 17.8% reduction; however, as the content further increased to 1%, the peak PWP rebounded to 3.90 kPa. The underlying mechanism for this phenomenon is that at the optimal content (0.75%), fibers construct a uniform and interconnected three-dimensional network within the soil matrix. These fiber bundles not only serve as "reinforcement" to constrain soil shear deformation but also function as efficient "micro-drainage channels", accelerating the dissipation of excess PWP and thereby maintaining it at a minimum level. Conversely, at insufficient fiber contents (<0.75%), such as in the 0.5% group, the fiber

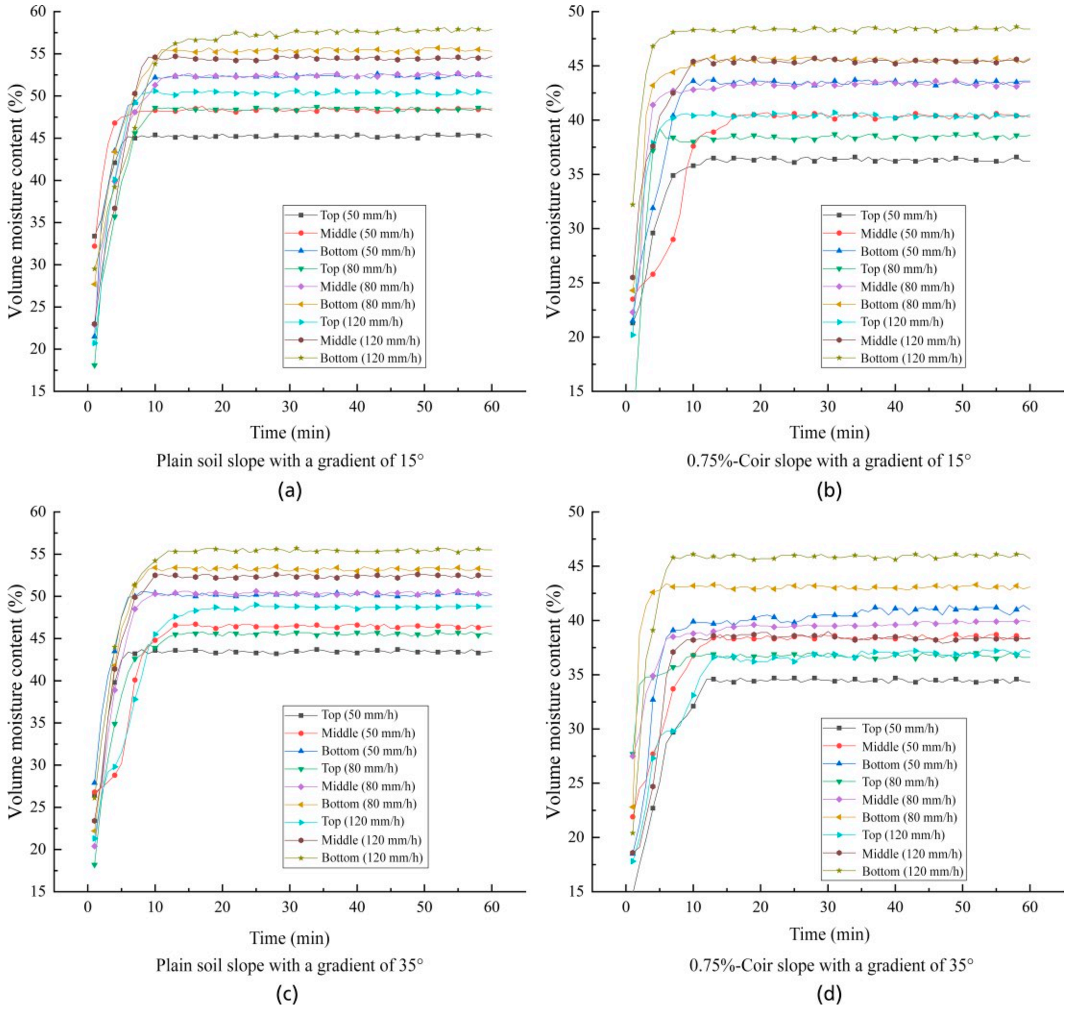

Figure 4. Variation curves of VWC in CF-reinforced soil slopes under different rainfall intensities

network is sparse and discontinuous, hindering the formation of continuous hydraulic pathways and resulting in a PWP reduction magnitude (approximately 5%∼8%) significantly lower than that of the optimal group. On the other hand, when fiber content is excessive (>0.75%), although fiber quantity increases, the resulting excessive porosity causes a sharp rise in permeability where the infiltration rate exceeds the drainage capacity. Furthermore, due to the inherent hydrophilicity and hollow lumen structure of CF, a "sponge effect" at high fiber contents enhances the water-holding capacity of the soil, causing entrapped moisture to be retained and leading to a rebound in PWP relative to the optimal state. In summary, a 0.75% CF content achieves an optimal synergistic balance between enhancing drainage capacity and controlling water retention characteristics, thereby maximally preserving the effective stress and stability of the slope under rainfall conditions.

4.2. Influence of PP on the hydraulic characteristics of slopes

The content of PP significantly influences the variation in peak VWC of the slope. Figure 6 illustrates the changes in peak VWC at different locations within the PP-reinforced soil slope under rainfall conditions. Observations indicate that PP reduces the peak VWC more effectively than the unreinforced slope, with the order of influence being: plain soil slope >0.3% > 0.7% > 0.5% fiber content. Notably, the optimal dosage for PP to control VWC is 0.5%, which is lower than the 0.75% threshold identified for CF in Section 4.1. This discrepancy stems from their fundamentally different material properties: CF, due to their hollow tubular structure and hydrophilicity, require a higher dosage to form a connected drainage network and tend to absorb water themselves. In contrast, PP possess superior hydrophobicity and high dispersibility; thus, a lower dosage (0.5%) is sufficient to construct an efficient "non-wetting" three-dimensional network within the soil. This network not only enhances soil cohesion and mitigates rain splash erosion but also restricts moisture retention within the soil matrix through hydrophobic repulsion. Consequently, under identical conditions, the water content of PP-reinforced slopes is generally lower than that of coir-reinforced slopes, which exhibit a "sponge effect".

Under varying rainfall intensities, the VWC of PP-reinforced slopes changes as shown in Figure 7. As rainfall intensity increases, the peak VWC rises, and the

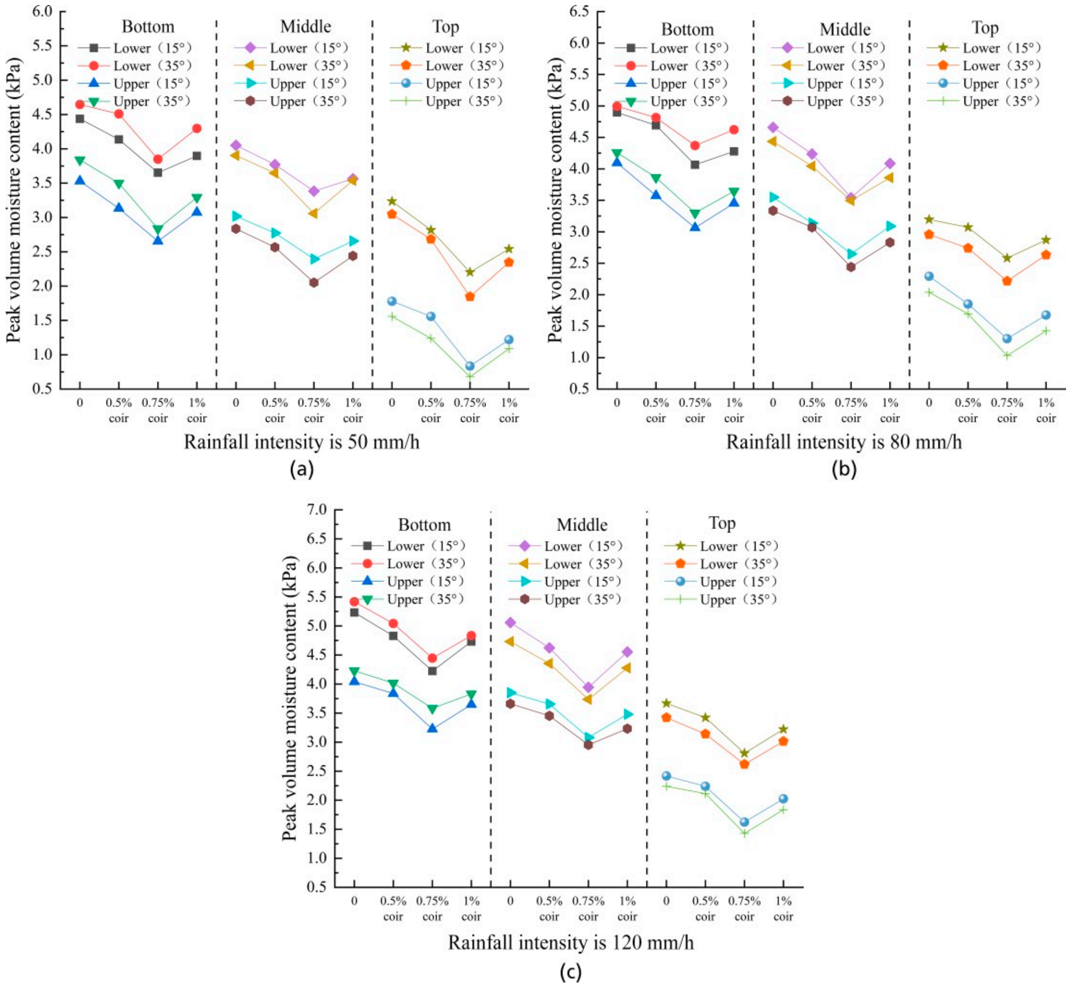

Figure 5. Variation in peak PWP of fiber-reinforced soil slopes at different CF content

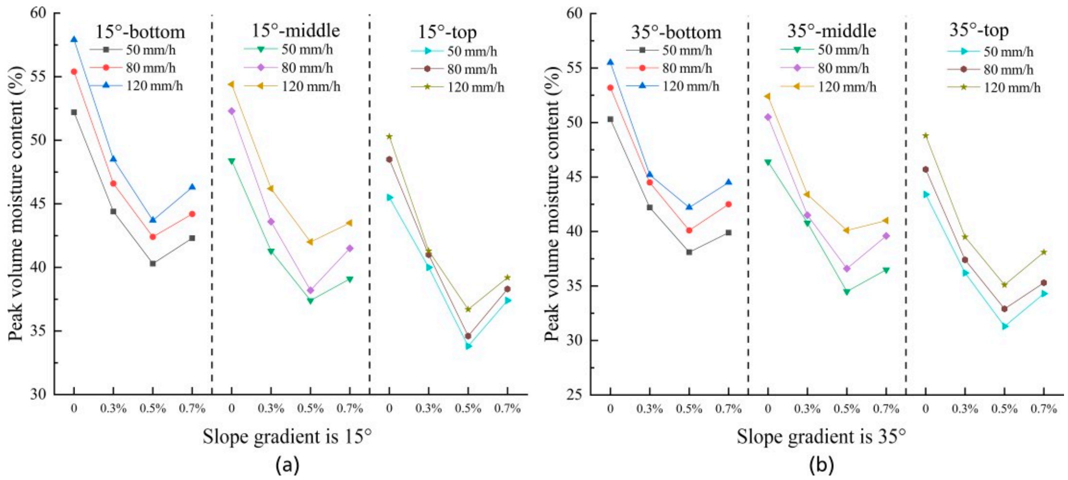

Figure 6. Variations in slope moisture content at different PP content

time to reach saturation decreases. At a 15° slope and 120 mm/h rainfall intensity, the VWC at all sections of the 0.5% PP-reinforced slope increased significantly. During this process, PP demonstrate a retardation mechanism distinct from that of CF: While CF delay infiltration primarily through hygroscopic buffering, PP significantly increase the tortuosity of seepage paths by dividing macropores with randomly distributed monofilaments. Furthermore, the hydrophobic nature of the fiber surfaces increases the contact angle at the soil-water interface, thereby physically impeding the rapid advancement of the wetting front. This mechanism facilitates uniform moisture redistribution and mitigates the risk of sudden, localized spikes in water content.

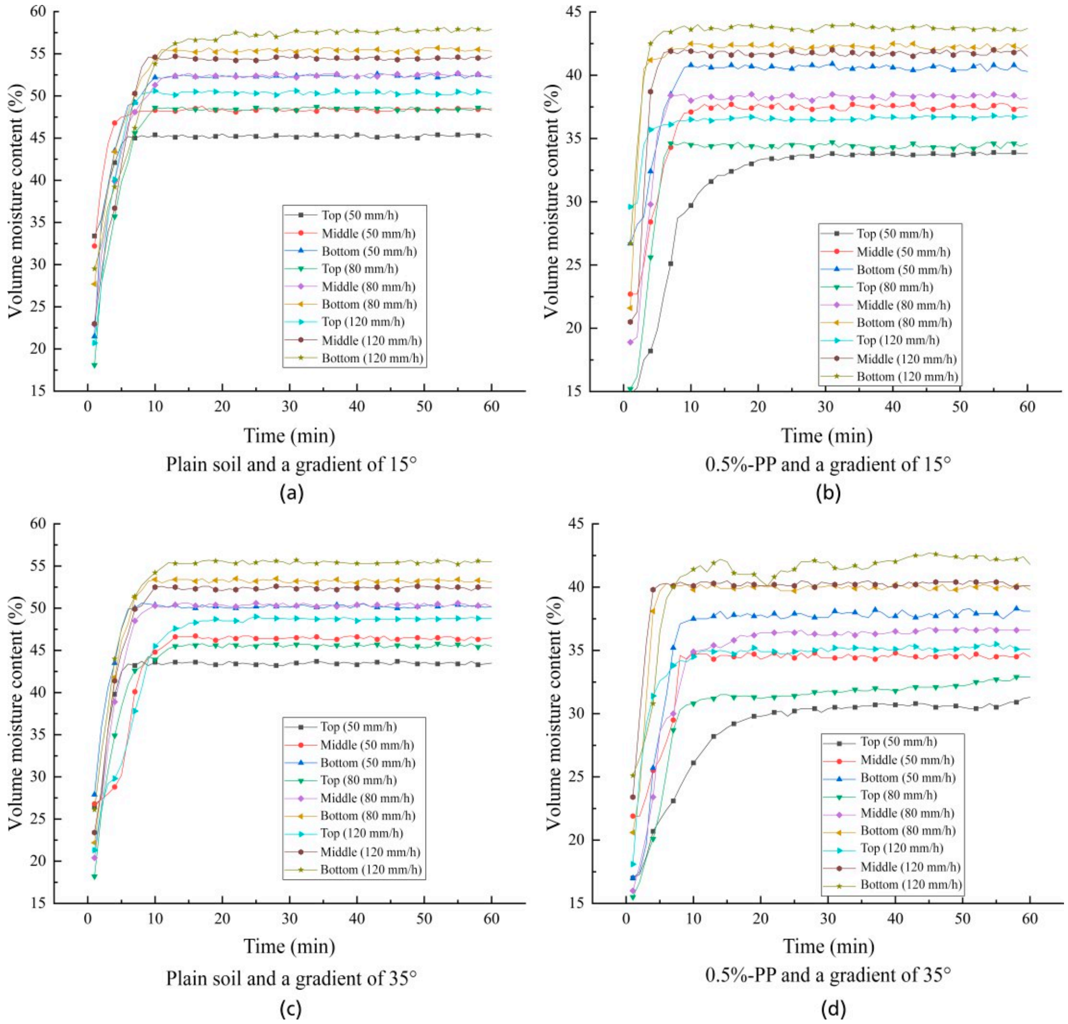

Figure 7. Variation curves of VWC in PP-reinforced soil slopes under different rainfall intensities

Figure 8 presents the variation in peak PWP within the reinforced slopes. The data show that peak PWP initially decreases and then increases with rising PP content, with the best performance observed at 0.5% dosage (a 26.9% reduction in the lower layer of the slope toe). Compared to CF, PP exhibit greater stability in reducing PWP. Although CF facilitate drainage, they are prone to swelling upon saturation, which may compress effective pore channels. Conversely, PP remain volumetrically stable in submerged environments. The interface formed between their smooth, hydrophobic surfaces and soil particles—while possessing slightly lower bonding force than the rough CF—creates faster "micro-drainage channels". This optimized pore structure not only shortens drainage paths but also avoids channel blockage caused by fiber swelling, thereby enabling more efficient dissipation of excess PWP under heavy rainfall and maintaining the effective stress of the slope.

4.3. Hydraulic characteristics of fiber-reinforced soil slopes using ryegrass

Fiber reinforcement not only enhances the mechanical properties of slopes but also optimizes their structural integrity and hydraulic behavior. Figure 9 illustrates the variations in the peak VWC of fiber-reinforced soil slopes under a rainfall intensity of 80 mm/h. The results demonstrate that incorporating fibers significantly reduces VWC. Specifically, the VWC reached its minimum value when the CF content was 0.75% and the PP content was 0.5%. Furthermore, the VWC of slopes reinforced with CF was consistently higher than that of slopes reinforced with PP, indicating that PP reinforcement is more effective in reducing slope saturation. As shown in Figure 9, under a rainfall intensity of 80 mm/h and a slope gradient of 15°, the overall VWC of the slope reinforced with 0.75% CF was 10.8% higher than that of the PP-reinforced slope.

This disparity is primarily attributed to the material properties of the fibers. CF possess high porosity and intrinsic hydrophilicity, which act to retain moisture within the soil matrix. In contrast, PP are chemically synthesized and exhibit strong hydrophobicity with low moisture retention, facilitating water drainage. Mechanistically, while the porous structure of CF facilitates initial infiltration, its high water absorption and binding effect with soil particles impede rapid drainage, resulting in a higher residual VWC. Conversely, the

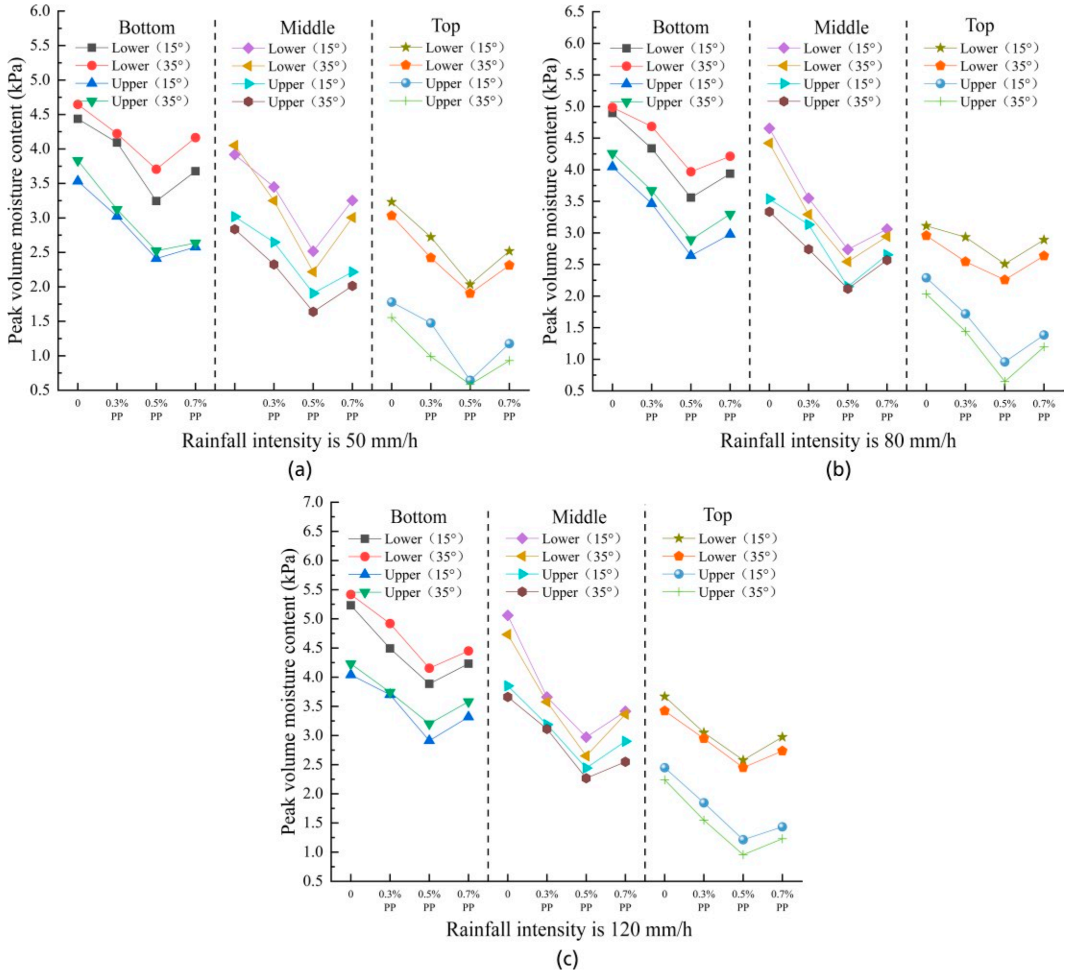

Figure 8. Variation in peak PWP within slope bodies at different PP content

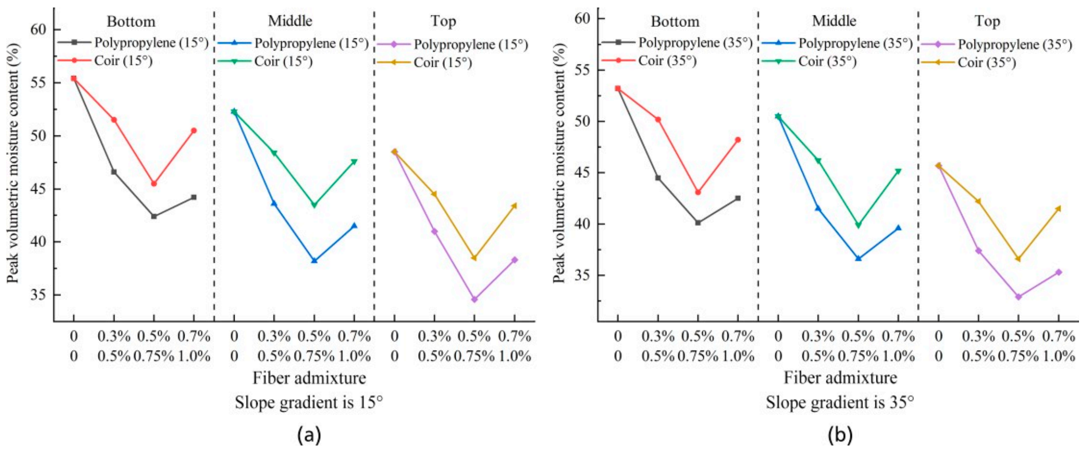

Figure 9. Fiber-reinforced soil slope VWC (rainfall intensity 80 mm/h)

hydrophobic nature of PP creates pathways that allow rainwater to drain swiftly through the pores, thereby minimizing water retention and maintaining a lower VWC within the soil mass.

The variation in peak VWC of the ryegrass-fiber reinforced soil slope body is shown in Figure 10. The graph indicates that the variation in VWC for reinforced slopes with different PP contents follows this sequence: ryegrass-unreinforced soil slope > ryegrass-0.3% PP reinforced soil slope > ryegrass-0.7% PP reinforced soil slope > ryegrass-0.5% PP reinforced soil slope. The variation in VWC across the reinforced soil slope body with different CF content was as follows: ryegrass-unreinforced soil slope > ryegrass-0.5% CF reinforced soil slope > ryegrass-1% CF reinforced soil slope > ryegrass-0.75% CF reinforced soil slope. Compared to the

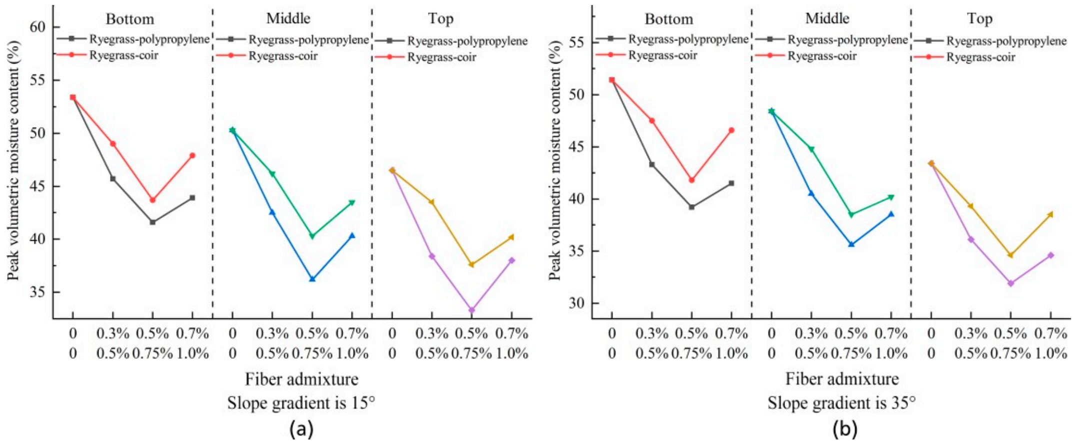

Figure 10. Ryegrass-fiber-reinforced soil slope VWC (rainfall intensity 80 mm/h)

ryegrass-unreinforced soil slope, the VWC at different sections of the ryegrass-0.5% PP reinforced soil slope decreased by 22.1% at the slope bottom, 28% at the slope middle, and 28.4% at the slope top. For the ryegrass-0.75% CF reinforced soil slope, VWC decreased by 18.2% at the bottom, 19.9% at the slope middle, and 19.1% at the slope top. This demonstrates that the drainage performance of the ryegrass-PP reinforced soil slope surpasses that of the ryegrass-CF reinforced soil slope.

As illustrated in Figures 11(a) and 11(b), under a rainfall intensity of 50 mm/h, the VWC of the ryegrass-reinforced slope with 0.75% CF decreased by 5.5%–10.4% compared to the fiber-only slope. In comparison, the slope reinforced with ryegrass and 0.5% PP exhibited a reduction of 4.7%∼8.2%. While planting ryegrass reduced the VWC in both slope types, the reduction was notably more pronounced in the CF-reinforced slope. This suggests that ryegrass exerts a stronger synergistic effect on water regulation in CF slopes than in PP slopes under low rainfall conditions. This phenomenon can be attributed to the higher porosity and inherent hydrophilicity of CF, which tend to retain moisture within the soil matrix. The introduction of ryegrass root systems, which possess both absorption and drainage functions, significantly enhances the otherwise limited drainage capacity of the coir-soil matrix. Conversely, PP, characterized by high dispersibility, form a denser soil structure with relatively uniform distribution. Since the PP-reinforced soil already possesses superior permeability due to the hydrophobic nature of the fibers, the marginal benefit of ryegrass roots on drainage is limited. Consequently, the regulatory impact of vegetation on VWC is less significant in PP-reinforced slopes compared to CF-reinforced slopes under moderate rainfall.

Figure 11 further demonstrates that the synergistic regulation of VWC by the "root-fiber-soil" system diminishes as rainfall intensity increases. However, under higher rainfall intensities, the synergistic efficacy of the ryegrass-PP system surpasses that of the ryegrass-coir system. The data reveal that the peak VWC of the ryegrass-coir slope consistently exceeds that of the ryegrass-PP slope. Specifically, at a rainfall intensity of 120 mm/h, the inclusion of ryegrass reduced the VWC by only 1.7%∼4.6% in the 0.75% CF slope, whereas it resulted in a 3.2%∼5.2% reduction in the 0.5% PP slope. This indicates that the synergy between ryegrass and CF weakens significantly under heavy rainfall. This is largely due to the inherent water absorption capacity of CF; as rainfall intensifies, water retention within the slope increases. Once soil moisture breaches a critical threshold, the slope's resistance to erosion weakens, increasing the risk of instability. In contrast, the ryegrass-PP slope maintains superior permeability because PP exhibit low water absorption, facilitating the rapid drainage of excess moisture. This keeps the soil VWC within a stable range. Consequently, as rainfall intensity increases, the ryegrass-PP reinforced slope demonstrates greater structural integrity and superior resistance to rainwater erosion compared to its CF counterpart.

Figure 12 illustrates the variation in PWP within the fiber-reinforced soil slope after planting ryegrass, under rainfall intensity of 80 mm/h. It can be observed that the variation in PWP within the PP-reinforced soil slope covered by ryegrass follows this sequence: ryegrass-undisturbed soil slope > ryegrass-0.3% PP-reinforced soil slope > ryegrass-0.7% PP-reinforced soil slope > ryegrass-0.5% PP-reinforced soil slope. For CF-reinforced slopes, the PWP variation follows: ryegrass-undisturbed soil slope > ryegrass-0.5% CF-reinforced slope > ryegrass-1% CF-reinforced slope > ryegrass-0.75% CF-reinforced slope. At identical gradients, the PWP within the slope body of the CF-reinforced soil slope remained higher than that of the PP-reinforced soil slope. Concurrently, as the gradient increased, PWP at both the mid-slope and crest of the fiber-reinforced soil slopes exhibited a decreasing trend, whilst PWP at the toe showed a certain degree of increase.

The variation trend of peak PWP within the ryegrass-fiber reinforced soil slope is illustrated in Figure 13. It can be observed that, compared to the fiber-reinforced soil slope, the peak PWP at different locations within the ryegrass-fiber-reinforced soil slope exhibits a significantly pronounced downward trend. This indicates that ryegrass possesses strong water absorption and drainage

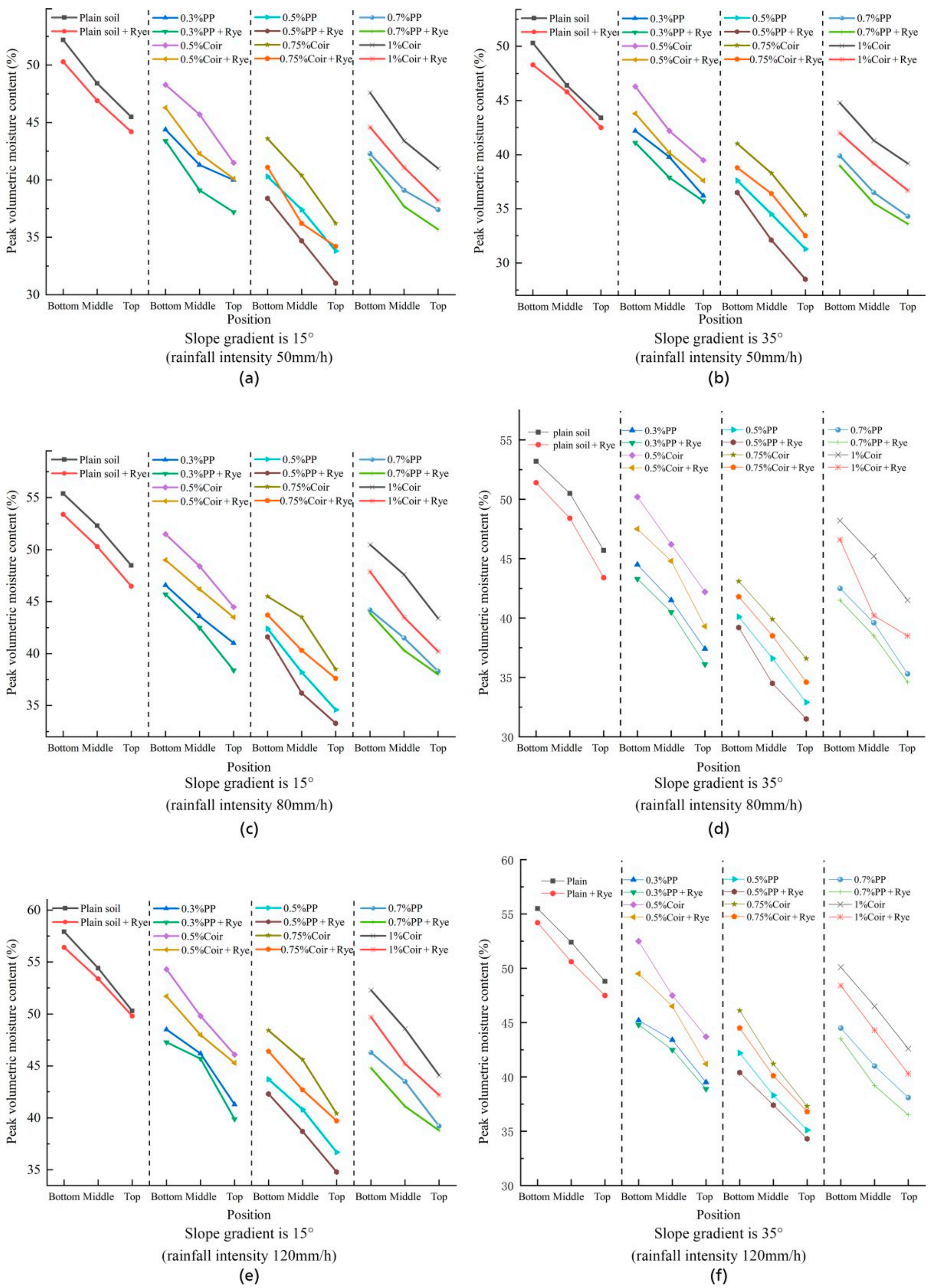

Figure 11. Variation in VWC of ryegrass-fiber reinforced soil slopes

capabilities, mitigating the rise in PWP during rainfall by reducing pore water accumulation within the slope body. Concurrently, the permeability of the slope body improved following ryegrass planting. This occurs because the ryegrass root system penetrates the soil during growth, forming numerous micro-pores that enhance the soil's permeable properties. Consequently, rainwater can drain more rapidly through the slope's pores, thereby slowing the rate of ascent of PWP within the slope body.

As illustrated in Figure 13, the peak PWP within the ryegrass-fiber reinforced slopes initially decreases and

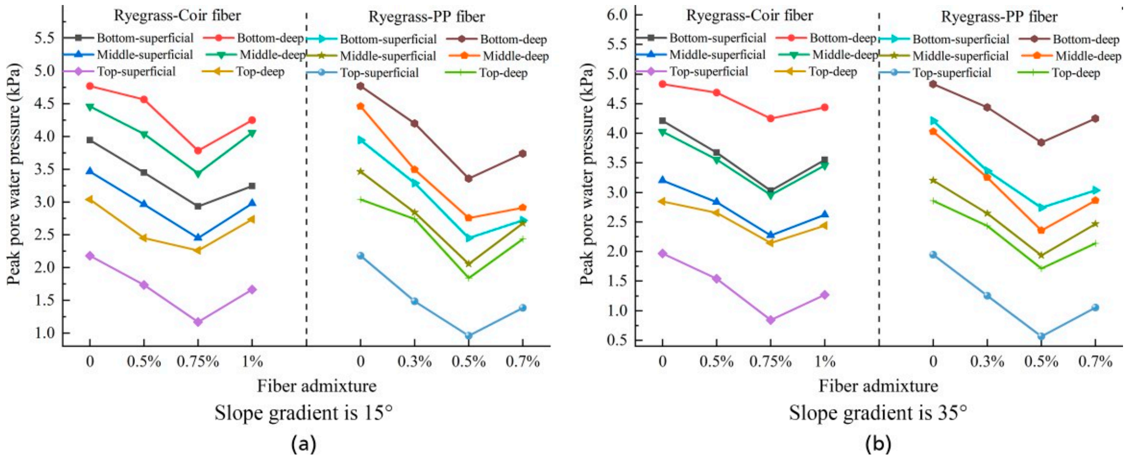

Figure 12. Variations in peak PWP in ryegrass-fiber reinforced soil (rainfall intensity 80 mm/h)

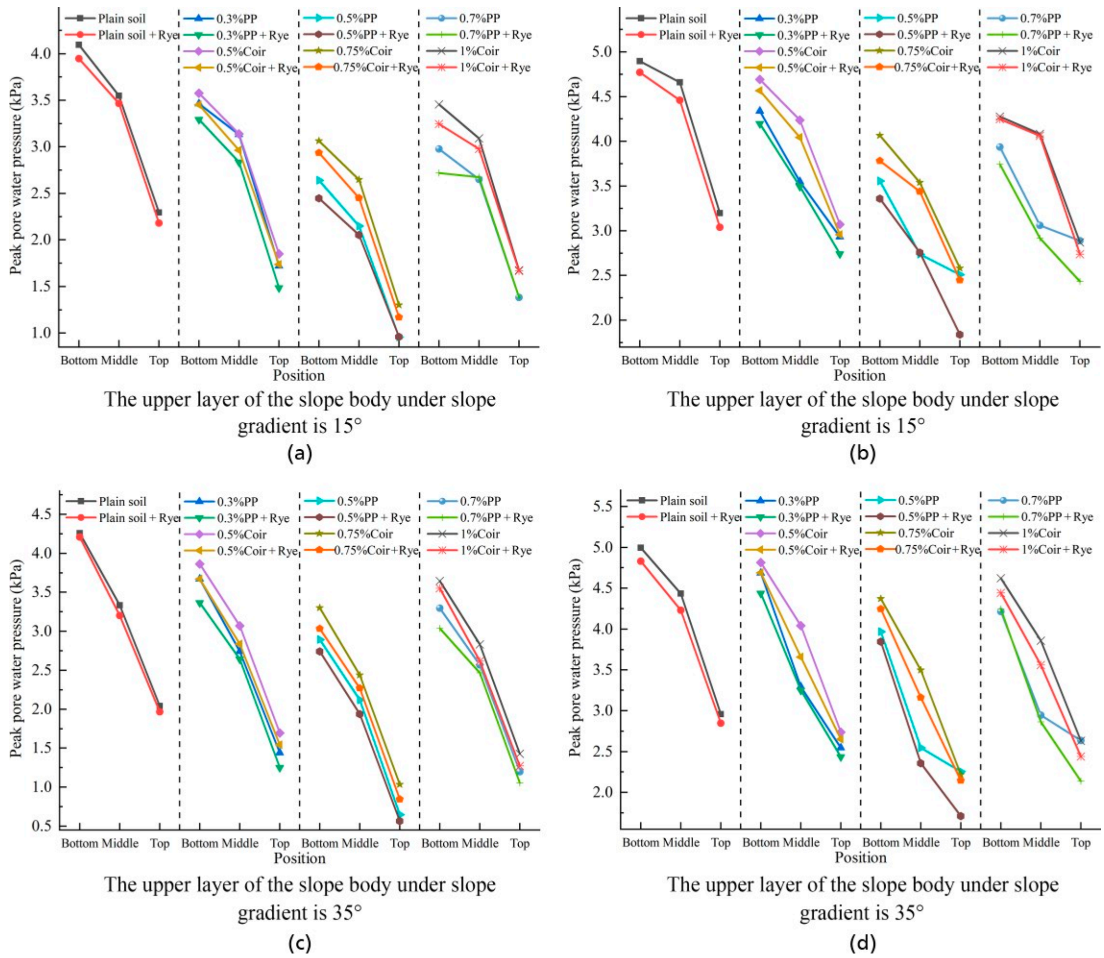

Figure 13. Variations in peak PWP within fiber-reinforced soil slopes of ryegrass (rainfall intensity 80 mm/h)

then increases with rising fiber content. The peak PWP reaches its minimum at fiber contents of 0.75% for CF and 0.5% for PP. Notably, at these optimal dosages, the peak PWP values in the ryegrass-reinforced systems are lower than those in the corresponding non-vegetated fiber-reinforced slopes. However, a comparison between the two fiber types reveals that at optimal dosages, the peak PWP in the ryegrass-CF slope remains higher than that in the ryegrass-PP slope. Under identical rainfall intensities, CF demonstrated a lower efficacy in reducing PWP compared to PP. Furthermore, the synergistic effect of ryegrass coupled with CF was less effective in optimizing hydraulic response than the ryegrass-PP combination.

This discrepancy suggests that the permeability of CF-reinforced slopes is inferior to that of PP-reinforced slopes, a phenomenon driven by material properties. Mechanistically, owing to the inherent hydrophilicity and high internal porosity of CF, they function as moisture reservoirs, absorbing and storing significant water. During rainfall, this characteristic enhances the water retention capacity of the slope, thereby sustaining PWP at elevated levels. In contrast, PP exhibit negligible water absorption and distribute uniformly within the soil matrix, effectively increasing macro-porosity and connectivity. Consequently, when combined with the root network of ryegrass, the ryegrass-PP system achieves superior drainage performance, facilitating rapid PWP dissipation compared to the ryegrass-CF system.

4.4. Research significance and research limitations

This study utilized systematic indoor model tests to profoundly reveal the non-linear hydraulic response mechanisms of the "vegetation-fiber-soil" three-phase composite medium under complex rainfall conditions. Theoretically, this work overcomes the bottlenecks of traditional studies that focus solely on either mechanical reinforcement or vegetation effects, constructing a novel framework for slope hydraulic regulation based on the hydrophilic and hydrophobic properties of fibers (Jiang *et al*. 2022b). Furthermore, it establishes differentiated criteria for "drainage-retention" functions: hydrophobic PP fibers utilize a "drainage synergy" effect to effectively reduce pore water pressure in heavy rainfall regions, whereas hydrophilic CF rely on a "sponge effect" to significantly enhance water retention capacity in arid regions. At the practical level, these data-driven results have profound practical implications for the engineering development of ecological slopes, particularly in improving soil stability and promoting sustainable vegetation growth. The quantitatively determined optimal dosage thresholds of 0.75% for CF and 0.5% for PP provide reliable parametric support for precise ecological design. The superior hydraulic regulation and soil consolidation efficacy of this "root-fiber" composite system fully confirm its application potential as a low-carbon, sustainable "green geotechnical" technology. It not only effectively overcomes the drawbacks of high costs and poor ecological benefits associated with traditional rigid supports but also offers an innovative solution for slope disaster prevention and resilience enhancement under extreme climatic conditions. Despite these advancements, this study inevitably has certain limitations: indoor model tests may introduce scale effects that are inconsistent with complex field conditions, and the relatively short observation period precludes a comprehensive evaluation of the long-term biodegradation process of fibers and its sustained impact on the in-service performance of the slope. Consequently, future research urgently needs to incorporate long-term in-situ field monitoring to further validate and optimize this ecological slope technology system.

5. CONCLUSION

Based on the indoor rainfall model tests considering the multi-factor coupling of rainfall intensity, slope gradient, and fiber wettability, the following conclusions are drawn regarding the hydraulic response mechanisms of fiber-reinforced and ryegrass-fiber composite slopes.

1. The wettability of fiber materials fundamentally dictates their hydraulic regulation functions. Hydrophobic PP facilitate a "drainage synergy" by reducing the resistance of water flow at the soil-fiber interface, effectively mitigating PWP accumulation. In contrast, hydrophilic fibers (CF) induce a "sponge effect" due to their hollow lumen structure, which enhances water retention but slows down drainage. The optimal dosages for balancing hydraulic regulation and mechanical reinforcement were identified as 0.5% for PP and 0.75% for CF.
2. The "root-fiber" composite system significantly outperforms single fiber reinforcement. The ryegrass-PP composite system demonstrated the superior drainage efficiency, reducing the peak VWC by up to 28.4% compared to the plain soil-ryegrass slope, whereas the ryegrass-CF system achieved a maximum reduction of 19.9%. This superiority is attributed to the complementary interaction where roots create preferential flow channels and hydrophobic fibers maintain these channels' patency, preventing moisture stagnation in the soil matrix.
3. This study establishes a criterion for material selection in ecological slope engineering based on climatic conditions. In regions with heavy rainfall where slope instability is driven by excess PWP, hydrophobic fibers (PP) are recommended to accelerate drainage and prevent softening. Conversely, in arid or semi-arid regions, hydrophilic fibers (CF) may be preferable to enhance soil moisture retention, thereby supporting vegetation survival during dry periods.
4. While this study provides theoretical insights into short-term hydraulic responses, it is limited by the scale effects of indoor models and the use of a single soil type. Future research should focus on:

Field Verification: Conducting large-scale in-situ tests to validate the "root-fiber" synergy under complex natural stratigraphy.

Long-term Durability: Investigating the biodegradation rate of natural CF and their time-dependent impact on slope stability.

Cyclic Response: Exploring the hydro-mechanical evolution of the composite system under long-term dry-wet and freeze-thaw cycles to assess the service life of ecological slopes.

FUNDING

The authors gratefully acknowledge the financial support from the National Key R&D Program of China

(Grant No. 2025YFE0123000), Natural Science Fund for Creative Groups of Hubei Province (Grant No. 2024AFA009), National Natural Science Foundation of China (Grant Nos. 52278347, 42407265), and the Outstanding Young and Middle-aged Scientific and Technological Innovation Team Project of Higher Education Institutions in Hubei Province (T2023006).

ABBREVIATIONS

CF	Coir fiber
CUC	Christiansen uniformity coefficient
CV	Coefficient of variation
FS	Full-scale linear error
PP	Hydrophobic polypropylene fiber
PWP	Pore water pressure
VWC	Volumetric water content

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